```
% Loads analysis
% Jacob Harris
% 10/2/26

format short

%Aircraft Datum = Nose of Aircraft
%All units in (in) or (lbf)

                                                    % 1G Static Case
%-------------------------------------------------------------------------------
--------------------------------%

        %Variables that can be changed
%-----------------------------------------%
%Horizontal Tail (assuming glued to fuselage)
L_HT = 4.71;                % in width of airfoil
t_HT = 0.57;                % in thickness of airfoil
ybar_HT_above = 0.11;       % in
A_HT = 1.81;                % in^2 area of airfoil
Ix_HT = 0.03;               % in^4
Iy_HT = 2.19;               % in^4
x_HT = 9;                   % in
NosetoLE_HT = 35.27;        % in
E_3Djoint = 220E3;          % psi
Ftu_3Djoint = 2000;         % psi
Fcu_3Djoint = 2000;         % psi
Fsu_3Djoint = 1000;         % psi

%Wing
L_w = 10.05;                % in width of airfoil
t_w = 0.96;                 % in thickness of airfoil
ybar_wing = 0.31;           % in
Ix_w = 0.28;                % in^4
Iy_w = 27.25;               % in^4
x_w = 50;                   % in
NosetoLE_w = 7.87;          % in

%Aircraft
AircraftWeight1G = 8.35;    % lbf
HorizontalCGfromLE = 2.8;   % in
E_epoxy = 400E3;            % psi
Ftu_Epoxy = 500;            % psi
Fcu_Epoxy = 5000;           % psi
Fsu_Epoxy = 200;            % psi

%Carbon Fiber Rods
numRods = 2;
d_rod_outer = 0.8;            % in
d_rod_inner = 0.7;          % in
E_rod = 19E6;               % psi
v = 0.33;
```

```
Ftu_rod = 250000;           % psi
Fcu_rod = 100000;           % psi
Fsu_rod = 6000;             % psi
%-----------------------------------------%

%Calculated Variables
c4_w = L_w/4;
c4wtoCG_w = HorizontalCGfromLE - c4_w;
c4_HT = L_HT/4;
CGtoc4_HT = NosetoLE_HT + c4_HT - NosetoLE_w - HorizontalCGfromLE;

%Reaction Forces
syms x
LHT = double(solve(0 == c4wtoCG_w*AircraftWeight1G - x*(CGtoc4_HT+c4wtoCG_w),x));
Lw = AircraftWeight1G - LHT;

%Wing Analysis
rodarea_outer = numRods*(pi()*d_rod_outer^2)/4;
rodarea_inner = numRods*(pi()*d_rod_inner^2)/4;
rodarea = rodarea_outer - rodarea_inner;
Ix_rod = numRods*(pi()/64)*(d_rod_outer^4 - d_rod_inner^4);
Q_rod = (d_rod_outer^3 - d_rod_inner^3)/12;
xlift_w = x_w/2;
RLw = -Lw/2;
MRLw = (Lw/2)*xlift_w;
c_rod = d_rod_outer/2;

tau_SW_max = (RLw*Q_rod)/(Ix_rod*(d_rod_outer-d_rod_inner));
tau_SW_avg = RLw/rodarea;                                   % psi
sigma_BW = MRLw*c_rod/Ix_rod;                               % psi

MS_SW_avg = (Fsu_rod/abs(tau_SW_avg))-1;                    % MS > 0
MS_SW_max = (Fsu_rod/abs(tau_SW_max))-1;                    % MS > 0
MS_BW_tension = (Ftu_rod/abs(sigma_BW))-1;                  % MS > 0
MS_BW_compression = (Fcu_rod/abs(sigma_BW))-1;              % MS > 0

%HT Analysis
xlift_HT = x_HT/2;
ybar_HT = t_HT/2;
A_HT_above = A_HT/2;
RLHT = -LHT/2;
MRLHT = (LHT/2)*xlift_HT;
c_HT = ybar_HT;
Q_HT = A_HT_above*ybar_HT_above;

tau_SHT_max = (RLHT*Q_HT)/(Ix_HT*L_HT);
tau_SHT_avg = RLHT/A_HT;                                    % psi
sigma_BWHT = MRLHT*c_HT/Ix_HT;                              % psi

MS_SHT_epoxy_max = (Fsu_Epoxy/abs(tau_SHT_max))-1;          % MS > 0
MS_BHT_epoxy_T = (Ftu_Epoxy/abs(sigma_BWHT))-1;             % MS > 0
MS_BHT_epoxy_C = (Fcu_Epoxy/abs(sigma_BWHT))-1;             % MS > 0
```

```
MS_SHT_3DJoint_max = (Fsu_3Djoint/abs(tau_SHT_max))-1;      % MS > 0
MS_BHT_3DJoint_T = (Ftu_3Djoint/abs(sigma_BWHT))-1;         % MS > 0
MS_BHT_3DJoint_C = (Fcu_3Djoint/abs(sigma_BWHT))-1;         % MS > 0

% Formatted results table
caseName = '1G Static Case';
headerFmt = '%-12s | %16s | %16s | %16s | %12s | %14s | %14s\n';
rowFmt    = '%-12s | %16.3f | %16.3f | %16.3f | %12.2f | %14.2f | %14.2f\n';
tableWidth = 118;
divider = repmat('-', 1, tableWidth);

titleText = [' ' caseName ' '];
leftPad = floor((tableWidth - length(titleText))/2);
rightPad = tableWidth - length(titleText) - leftPad;

fprintf('\n%s%s%s\n',repmat('-',1,leftPad), titleText, repmat('-',1,rightPad));
```

```
---------------------------------------------- 1G Static Case ----------------------------------------------
```

```
fprintf(headerFmt,'Material', 'Avg Shear (psi)', 'Max Shear (psi)', 'Bending
(psi)','Max Shear MS', '(T) Bending MS', '(C) Bending MS');
```

```
Material     |  Avg Shear (psi) |  Max Shear (psi) |    Bending (psi) | Max Shear MS | (T) Bending MS | (C) Bending
```

```
fprintf('%s\n', divider);
```

```
----------------------------------------------------------------------------------------------------------------
```

```
fprintf(rowFmt, 'CF Rods(Wing)',abs(tau_SW_avg),abs(tau_SW_max),
abs(sigma_BW),MS_SW_max, MS_BW_tension, MS_BW_compression);
```

```
CF Rods(Wing) |           17.524 |           34.944 |         2481.247 |       170.70 |          99.76 |          39
```

```
fprintf(rowFmt, 'Epoxy(HT)',abs(tau_SHT_avg),abs(tau_SHT_max),
abs(sigma_BWHT),MS_SHT_epoxy_max, MS_BHT_epoxy_T, MS_BHT_epoxy_C);
```

```
Epoxy(HT)    |           0.025 |           0.032 |           1.969 |      6163.45 |         252.98 |        2538
```

```
fprintf(rowFmt, '3D Joint(HT)',abs(tau_SHT_avg),abs(tau_SHT_max),
abs(sigma_BWHT),MS_SHT_3DJoint_max, MS_BHT_3DJoint_T, MS_BHT_3DJoint_C);
```

```
3D Joint(HT) |           0.025 |           0.032 |           1.969 |     30821.25 |        1014.91 |        1014
```

```
fprintf('%s\n', divider);
```

```
----------------------------------------------------------------------------------------------------------------
```

```
fprintf('MS is dimensionless; positive values indicate passing.\n');
```

```
MS is dimensionless; positive values indicate passing.
```

```
fprintf('HT: Horizontal Tail.\n');
```

```
HT: Horizontal Tail.
```

```
fprintf('\n\n');
%-------------------------------------------------------------------------------
--------------------------------%
```

```
                                                    % 2G static Case
%-------------------------------------------------------------------------------
--------------------------------%

%Aircraft
AircraftWeight2G = AircraftWeight1G*2;    % lbf

%Reaction Forces
syms x
LHT2 = double(solve(0 == c4wtoCG_w*AircraftWeight2G - x*(CGtoc4_HT+c4wtoCG_w),x));
Lw2 = AircraftWeight2G - LHT2;

%Wing Analysis
RLw2 = -Lw2/2;
MRLw2 = (Lw2/2)*xlift_w;

tau_SW_max2 = (RLw2*Q_rod)/(Ix_rod*(d_rod_outer-d_rod_inner));  % psi
tau_SW_avg2 = RLw2/rodarea;                                     % psi
sigma_BW2 = MRLw2*c_rod/Ix_rod;                                 % psi

MS_SW_avg2 = (Fsu_rod/abs(tau_SW_avg2))-1;                      % MS > 0
MS_SW_max2 = (Fsu_rod/abs(tau_SW_max2))-1;                      % MS > 0
MS_BW_tension2 = (Ftu_rod/abs(sigma_BW2))-1;                    % MS > 0
MS_BW_compression2 = (Fcu_rod/abs(sigma_BW2))-1;                % MS > 0

%Horizontal Tail Analysis
RLHT2 = -LHT2/2;
MRLHT2 = (LHT2/2)*xlift_HT;

tau_SHT_max2 = (RLHT2*Q_HT)/(Ix_HT*L_HT);
tau_SHT_avg2 = RLHT2/A_HT;                                    % psi
sigma_BWHT2 = MRLHT2*c_HT/Ix_HT;                              % psi

MS_SHT_epoxy_max2 = (Fsu_Epoxy/abs(tau_SHT_max2))-1;           % MS > 0
MS_BHT_epoxy_T2 = (Ftu_Epoxy/abs(sigma_BWHT2))-1;              % MS > 0
MS_BHT_epoxy_C2 = (Fcu_Epoxy/abs(sigma_BWHT2))-1;              % MS > 0
MS_SHT_3DJoint_max2 = (Fsu_3Djoint/abs(tau_SHT_max2))-1;       % MS > 0
MS_BHT_3DJoint_T2 = (Ftu_3Djoint/abs(sigma_BWHT2))-1;          % MS > 0
MS_BHT_3DJoint_C2 = (Fcu_3Djoint/abs(sigma_BWHT2))-1;          % MS > 0

% Formatted results table
caseName = '2G Static Case';
headerFmt = '%-12s | %16s | %16s | %16s | %12s | %14s | %14s\n';
rowFmt    = '%-12s | %16.3f | %16.3f | %16.3f | %12.2f | %14.2f | %14.2f\n';
tableWidth = 118;
divider = repmat('-', 1, tableWidth);

titleText = [' ' caseName ' '];
```

```
leftPad = floor((tableWidth - length(titleText))/2);
rightPad = tableWidth - length(titleText) - leftPad;

fprintf('\n%s%s%s\n',repmat('-',1,leftPad), titleText, repmat('-',1,rightPad));
```

```
---------------------------------------------- 2G Static Case ----------------------------------------------
```

```
fprintf(headerFmt,'Material', 'Avg Shear (psi)', 'Max Shear (psi)', 'Bending
(psi)','Max Shear MS', '(T) Bending MS', '(C) Bending MS');
```

```
Material     |  Avg Shear (psi) |  Max Shear (psi) |    Bending (psi) | Max Shear MS | (T) Bending MS | (C) Bending
```

```
fprintf('%s\n', divider);
```

```
-------------------------------------------------------------------------------------------------------------
```

```
fprintf(rowFmt, 'CF Rods(Wing)',abs(tau_SW_avg2),abs(tau_SW_max2),
abs(sigma_BW2),MS_SW_max2, MS_BW_tension2, MS_BW_compression2);
```

```
CF Rods(Wing) |          35.048 |          69.888 |        4962.493 |        84.85 |          49.38 |          19
```

```
fprintf(rowFmt, 'Epoxy(HT)',abs(tau_SHT_avg2),abs(tau_SHT_max2),
abs(sigma_BWHT2),MS_SHT_epoxy_max2, MS_BHT_epoxy_T2, MS_BHT_epoxy_C2);
```

```
Epoxy(HT)    |          0.051 |          0.065 |           3.937 |      3081.22 |        125.99 |        1268
```

```
fprintf(rowFmt, '3D Joint(HT)',abs(tau_SHT_avg2),abs(tau_SHT_max2),
abs(sigma_BWHT2),MS_SHT_3DJoint_max2, MS_BHT_3DJoint_T2, MS_BHT_3DJoint_C2);
```

```
3D Joint(HT) |          0.051 |          0.065 |           3.937 |     15410.12 |        506.96 |         506
```

```
fprintf('%s\n', divider);
```

```
-------------------------------------------------------------------------------------------------------------
```

```
fprintf('MS is dimensionless; positive values indicate passing.\n');
```

```
MS is dimensionless; positive values indicate passing.
```

```
fprintf('\n\n');
%-------------------------------------------------------------------------------
--------------------------------%




                                                    % 2G Landing Static Case
%-------------------------------------------------------------------------------
--------------------------------%

        %Variables that can be changed
%-----------------------------------------%

%Landing Gears
numLG = 4;
z_FLG = 4.95;           % in front landing gear from nose
z_BLG = 14.95;          % in back landing gear from nose
```

```
d_LG = 0.875;            % in
x_FLG = 4.36;           % in
x_BLG = 4.56;           % in
theta_BLG = 35.83;      % deg

%-----------------------------------------%

%Calculated Variables
%AircraftWeight2G = AircraftWeight1G;
FLGtoCG = (NosetoLE_w + HorizontalCGfromLE) - z_FLG;
CGtoBLG = z_BLG - (NosetoLE_w + HorizontalCGfromLE);
A_LG = (pi()*d_LG^2)/4;
Ix_LG = pi()*d_LG^4/64;
Q_LG = d_LG^3/12;
c_LG = d_LG/2;

%Reaction Forces
syms x
LBLG = double(solve(0 == FLGtoCG*AircraftWeight2G - x*(CGtoBLG+FLGtoCG),x));
LFLG = AircraftWeight2G - LBLG;

%Front Landing Gear Analysis
RFLG = -LFLG/2;
MRFLG = (LFLG/2)*x_FLG;

tau_FLG_max = (RFLG*Q_LG)/(Ix_LG*d_LG);  % psi
tau_FLG_avg = RFLG/A_LG;                 % psi
sigma_BFLG = MRFLG*c_LG/Ix_LG;           % psi

MS_SFLG_epoxy_max = (Fsu_Epoxy/abs(tau_FLG_max))-1;          % MS > 0
MS_BFLG_epoxy_T = (Ftu_Epoxy/abs(sigma_BFLG))-1;             % MS > 0
MS_BFLG_epoxy_C = (Fcu_Epoxy/abs(sigma_BFLG))-1;             % MS > 0
MS_SFLG_3DJoint_max = (Fsu_3Djoint/abs(tau_FLG_max))-1;      % MS > 0
MS_BFLG_3DJoint_T = (Ftu_3Djoint/abs(sigma_BFLG))-1;         % MS > 0
MS_BFLG_3DJoint_C = (Fcu_3Djoint/abs(sigma_BFLG))-1;         % MS > 0

%Back Landing Gear Analysis
RBLG = -LBLG/2;
MRBLG = (LBLG/2)*x_BLG;
PBLG_local = RBLG*cos(theta_BLG*pi()/180);
VBLG_local = RBLG*sin(theta_BLG*pi()/180);

sigma_BLG_local = PBLG_local/A_LG;              % psi
tau_BLG_max = (VBLG_local*Q_LG)/(Ix_LG*d_LG);   % psi
tau_BLG_avg = VBLG_local/A_LG;                  % psi
sigma_BBLG = MRBLG*c_LG/Ix_LG;                  % psi

MS_ABLG_epoxy_max_T = (Ftu_Epoxy/abs(sigma_BLG_local))-1;
MS_ABLG_epoxy_max_C = (Fcu_Epoxy/abs(sigma_BLG_local))-1;
MS_SBLG_epoxy_max = (Fsu_Epoxy/abs(tau_BLG_max))-1;          % MS > 0
MS_BBLG_epoxy_T = (Ftu_Epoxy/abs(sigma_BBLG))-1;             % MS > 0
MS_BBLG_epoxy_C = (Fcu_Epoxy/abs(sigma_BBLG))-1;             % MS > 0
```

```
MS_ABLG_3DJoint_max_T = (Ftu_3Djoint/abs(sigma_BLG_local))-1;
MS_ABLG_3DJoint_max_C = (Fcu_3Djoint/abs(sigma_BLG_local))-1;
MS_SBLG_3DJoint_max = (Fsu_3Djoint/abs(tau_BLG_max))-1;      % MS > 0
MS_BBLG_3DJoint_T = (Ftu_3Djoint/abs(sigma_BBLG))-1;         % MS > 0
MS_BBLG_3DJoint_C = (Fcu_3Djoint/abs(sigma_BBLG))-1;         % MS > 0

% Formatted results table
caseName = '2G Landing Case';
headerFmt = ['%-14s | %15s | %15s | %15s | ''%12s | %12s | %12s | %14s | %14s\n'];
rowFmt = ['%-14s | %15.3f | %15.3f | %15.3f | ''%12s | %12s | %12.2f | %14.2f |
%14.2f\n'];
headerText = sprintf(headerFmt,'Material', 'Avg Shear (psi)', 'Max Shear (psi)', ...
    'Bending (psi)', '(T) Axial MS', '(C) Axial MS','Max Shear MS', '(T) Bending
MS', '(C) Bending MS');

tableWidth = length(headerText) - 1;
divider = repmat('-', 1, tableWidth);

titleText = [' ' caseName ' '];
leftPad = floor((tableWidth - length(titleText))/2);
rightPad = tableWidth - length(titleText) - leftPad;

fprintf('\n%s%s%s\n',repmat('-',1,leftPad), titleText, repmat('-',1,rightPad));
```

```
---------------------------------------------------------------- 2G Landing Case ----------------------------------
```

```
fprintf('%s', headerText);
```

```
Material       | Avg Shear (psi) | Max Shear (psi) |   Bending (psi) | '(T) Axial MS | (C) Axial MS | Max Shear MS
```

```
fprintf('%s\n', divider);
```

```
-------------------------------------------------------------------------------------------------------------------
```

```
% Front landing gear
fprintf(rowFmt, 'Epoxy(FLG)',abs(tau_FLG_avg), abs(tau_FLG_max), abs(sigma_BFLG),
...
    'N/A', 'N/A', MS_SFLG_epoxy_max, MS_BFLG_epoxy_T, MS_BFLG_epoxy_C);
```

```
Epoxy(FLG)     |           5.943 |           7.924 |         236.915 | '         N/A |          N/A |         24.24
```

```
fprintf(rowFmt, '3D Joint(FLG)',abs(tau_FLG_avg), abs(tau_FLG_max),
abs(sigma_BFLG), ...
    'N/A', 'N/A', MS_SFLG_3DJoint_max, MS_BFLG_3DJoint_T, MS_BFLG_3DJoint_C);
```

```
3D Joint(FLG)  |           5.943 |           7.924 |         236.915 | '         N/A |          N/A |        125.19
```

```
% Back landing gear
fprintf(rowFmt, 'Epoxy(BLG)', abs(tau_BLG_avg), abs(tau_BLG_max), abs(sigma_BBLG),
...
    sprintf('%.2f', MS_ABLG_epoxy_max_T),sprintf('%.2f', MS_ABLG_epoxy_max_C), ...
    MS_SBLG_epoxy_max, MS_BBLG_epoxy_T, MS_BBLG_epoxy_C);
```

```
Epoxy(BLG)     |           4.650 |           6.199 |         331.149 | '        76.64 |        775.43 |         31.26
```

```
fprintf(rowFmt, '3D Joint(BLG)', abs(tau_BLG_avg), abs(tau_BLG_max),
abs(sigma_BBLG), ...
    sprintf('%.2f', MS_ABLG_3DJoint_max_T),sprintf('%.2f', MS_ABLG_3DJoint_max_C),
...
    MS_SBLG_3DJoint_max, MS_BBLG_3DJoint_T, MS_BBLG_3DJoint_C);
```

```
3D Joint(BLG)  |          4.650 |          6.199 |        331.149 | '      309.57 |       309.57 |       160.30
```

```
fprintf('%s\n', divider);
```

```
-----------------------------------------------------------------------------------------------------------------
```

```
fprintf('MS is dimensionless; positive values indicate passing.\n');
```

```
MS is dimensionless; positive values indicate passing.
```

```
fprintf('N/A: no axial load included in the front-gear model.\n');
```

```
N/A: no axial load included in the front-gear model.
```

```
fprintf('FLG: Front Landing Gear\n');
```

```
FLG: Front Landing Gear
```

```
fprintf('BLG: Back Landing Gear\n');
```

```
BLG: Back Landing Gear
```

```
%-------------------------------------------------------------------------------
--------------------------------%
```